

Embedding a Family of Two Disjoint Multi-dimensional Meshes into Locally Twisted Cubes *

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Abstract

Let LTQ_n denote the n -dimensional locally twisted cubes. This paper deals with the problem of how to embed a family of two disjoint multi-dimensional meshes into locally twisted cubes. We develop the following embeddings: for $n \geq 3$ and $2 \leq k \leq n$, two disjoint meshes with size $\underbrace{2 \times 2 \times \cdots \times 2}_{k-1} \times 2^{n-k}$ can be embedded into LTQ_n with unit dilation, unit expansion, and congestion-free. The results obtained are optimal in the sense that the dilations, expansions and congestions of all the embeddings are equal to 1.

1 Introduction

Interconnection networks are usually modeled as undirected simple graphs $G = (V, E)$, where the node set V and the edge set E represent the sets of processing elements and communication channels, respectively. One of the central issues in evaluating the efficiency of an interconnection network is to consider the ability of graph embedding, which is a technique used for processor allocation and architecture simulation [1, 2, 20]. For the former, a parallel algorithm with structured task graph (i.e., a guest graph) is mapped into a network (i.e., a host graph) such that the algorithm can be executed efficiently. For the latter, a guest architecture is mapped into another host architecture for simulation.

Formally, *graph embedding* can be defined as follows: given two graphs $G = (V_G, E_G)$ and $H = (V_H, E_H)$, an embedding from the guest

graph G to the host graph H is an injective mapping $\psi : V_G \rightarrow V_H$. Under this mapping, it is clear that $|V_H| \geq |V_G|$ and an edge of G corresponds to a path of H . The performance of an embedding can be measured by the three parameters: *dilation*, *expansion* and *congestion*. The dilation is expressed by the maximum length of a path in H that is mapped by an edge of G . Thus, the smaller dilation of an embedding has the shorter communication delay when H simulates G . The expansion is defined as the ratio $|V_H|/|V_G|$ for measuring the processor utilization. Thus, the smaller expansion of an embedding is more efficient in processor utilization when H simulates G . The congestion is defined to be the maximum number of times that an edge of H is contained in the paths corresponding to the edges of G . In particular, if the embedding congestion is 1, we may interchangeably adopt the term *congestion-free* to reflect the fact that no two edges of G share a same link in H .

Graph embedding have been well-studied for guest graphs as meshes, trees, cycles, paths, etc. (see, e.g. [3–13, 15–19, 21–24, 26]), because they are the common data structures or interconnection structures used in parallel computing. The n -dimensional locally twisted cube, denoted as LTQ_n , is a variant of the n -dimensional hypercube, but has some better features than the hypercube. One advantage of LTQ_n is that the diameter is only about half of the diameter of an n -dimensional hypercube. For more properties of LTQ_n , the reader can refer to [3, 13–15, 18, 19, 25–27]. Graph embeddings in locally twisted cubes have been studied by considering that the guest graphs are paths [19], cycles [3, 15, 26], and trees [13, 18]. However, to our knowledge, so far the problem of embedding a family of disjoint meshes into a locally twisted cube has not yet to be investigated in the literature. In this pa-

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per, we will consider this issue in locally twisted cubes and prove the following result: for $n \geq 3$ and each $2 \leq k \leq n$, two disjoint meshes each of size $\underbrace{2 \times 2 \times \cdots \times 2}_{k-1} \times 2^{n-k}$ can be embedded into

LTQ_n with unit dilation, unit expansion, and congestion-free. In particular, if $k = n$ or $k = n-1$, the above embedding shows that LTQ_n consists of two disjoint copies of $(n-1)$ -dimensional hypercube Q_{n-1} .

The rest of this paper is organized as follows. Section 2 formally gives the definition of locally twisted cubes and introduces some basic properties. Section 3 establishes the main result. The final section contains our concluding remarks.

2 Preliminaries

Let $\oplus$ represent the modulo 2 addition. For $n \geq 2$, the n -dimensional locally twisted cube LTQ_n , proposed first by Yang et al. [25], is a graph with $\{0,1\}^n$ as its node set, and two nodes $x = x_{n-1}x_{n-2}\cdots x_0$ and $y = y_{n-1}y_{n-2}\cdots y_0$ are adjacent in LTQ_n if and only if either

- (1) there is an integer $i \in \{2, 3, \dots, n-1\}$ such that $x_i = \bar{y}_i$ and $x_{i-1} = y_{i-1} \oplus x_0$, and $x_j = y_j$ for all remaining bits, or
- (2) there is an integer $i \in \{0, 1\}$ such that $x_i = \bar{y}_i$, and $x_j = y_j$ for all remaining bits.

If one of the above conditions is fulfilled, then y is called the i -neighbor of x , denoted by $N_i(x)$. Also, (x, y) is called an i -dimensional edge of LTQ_n . Figure 1 shows the graphs LTQ_3 and LTQ_4 , respectively. The locally twisted cube can be equivalently defined with the following recursively fashion:

- (1) LTQ_2 is a graph consisting of four nodes labeled with 00, 01, 10, and 11, respectively, connected by four edges (00,01), (00,10), (01,11), and (10,11).
- (2) For $n \geq 3$, LTQ_n is constructed from two disjoint copies of LTQ_{n-1} according to the following steps. Let 0LTQ_n (respectively, 1LTQ_n) denote the graph obtained by prefixing the label of each node in one copy of LTQ_{n-1} with 0 (respectively, 1). Each node $0x_{n-2}x_{n-3}\cdots x_0$ in 0LTQ_n is connected with the node $1(x_{n-2} \oplus x_0)x_{n-3}\cdots x_0$ in 1LTQ_n by an edge.

From the above definition, it is clear that LTQ_n is an n -regular graph, and the binary strings of

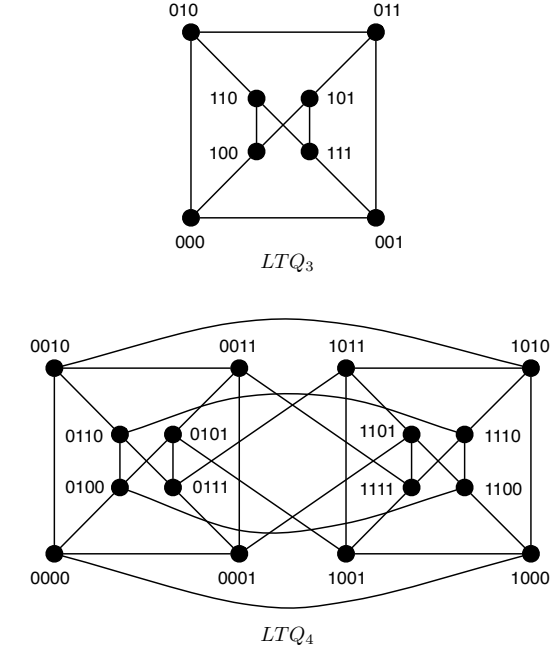


Figure 1: Locally twisted cubes LTQ_3 and LTQ_4 .

any two adjacent nodes in LTQ_n differ in at most two successive bits. For notation convenience, if a node x is contained in LTQ_n , we simply write $x \in LTQ_n$. The following proposition is easy to be checked from the definition of LTQ_n .

Proposition 1 For $x, y \in LTQ_n$ and $0 \leq i, j \leq n-1$, the following statements are true:

- (1) $N_i(x) = y$ if and only if $N_i(y) = x$. In particular, $N_i(N_i(x)) = x$.
- (2) $N_i(N_j(x)) = N_j(N_i(x))$ for $i \neq j$.

For LTQ_n , we use notation iLTQ_n and LTQ_n^j for $i, j \in \{0, 1\}$ to denote the subgraphs of LTQ_n induced by the set of nodes with i as their first bit and j as their last bit, respectively. Similarly, we may use ${}^iLTQ_n^j$ to denote the subgraph of LTQ_n with nodes having restricted prefix label and suffix label, simultaneously. By definition, each subgraph iLTQ_n is isomorphic to LTQ_{n-1} under the isomorphism $\varphi(x) = x'$, where $x = ix_{n-2}\cdots x_0 \in {}^iLTQ_n$ and $x' = x_{n-2}\cdots x_0 \in LTQ_{n-1}$. Oppositely, LTQ_n^j is not isomorphic to LTQ_{n-1} . In [26], Yang et al. proved that LTQ_n is Hamiltonian-connected, i.e., every two nodes are connected by a Hamiltonian path (see also [19] for a stronger property). However, this property does not hold for the subgraph LTQ_n^j . For example, there is no path of length 3 between 000 and

110 in LTQ_3^0 . If P and Q are two paths with a common end-node, we denote $P + Q$ to mean the concatenation of P and Q . The following lemma shows that every node and its $(n-1)$ -neighbor are connected by a Hamiltonian path in the subgraph LTQ_n^j .

Lemma 2 *For $n \geq 3$ and $j \in \{0, 1\}$, there is a Hamiltonian path connecting node $x \in LTQ_n^j$ and its $(n-1)$ -neighbor.*

Proof. We prove the lemma by induction on n . Since LTQ_3^0 and LTQ_3^1 are 4-cycles, it is easy to see that the lemma holds for $n = 3$. Suppose $n \geq 4$ and the lemma holds for $n-1$. Let x be any node in the subgraph LTQ_n^j and let $y = N_{n-1}(x)$. Without loss of generality assume that $x \in {}^0LTQ_n^j$ and $y \in {}^1LTQ_n^j$. The aim is to construct a Hamiltonian path connecting x and y in LTQ_n^j . Since ${}^iLTQ_n^j$ is isomorphic to LTQ_{n-1}^j for $i \in \{0, 1\}$, by induction hypothesis, there is a Hamiltonian path connecting every node and its $(n-2)$ -neighbor in ${}^iLTQ_n^j$. Let $x' = N_{n-2}(x)$ and P be a Hamiltonian path connecting x and x' in ${}^0LTQ_n^j$. Also, let $y' = N_{n-2}(y)$ and Q be a Hamiltonian path connecting y' and y in ${}^1LTQ_n^j$. By Proposition 1, $y' = N_{n-2}(y) = N_{n-2}(N_{n-1}(x)) = N_{n-1}(N_{n-2}(x)) = N_{n-1}(x')$. Thus, (x', y') is an $(n-1)$ -dimensional edge. As a consequence, $P + (x', y') + Q$ is the desired Hamiltonian path that connects x and y in LTQ_n^j . $\square$

For $x \in LTQ_n^j$, let HP_x denote the Hamiltonian path constructed in Lemma 2. From the constructing rule, we will see that nodes and edges in HP_x are appeared in a recursively fashion. Let $HP_x = (x = x_1, x_2, \dots, x_p)$ where $p = 2^{n-1}$ and $x_p = N_{n-1}(x)$. Suppose that (x_i, x_{i+1}) is an λ_i -dimensional edge for $1 \leq i \leq p-1$. It is easy to check that $\lambda_i = \lambda_{p-i}$. Moreover,

$$\begin{cases} \lambda_{2k+1} = 1 & \text{for } 0 \leq k \leq \frac{p}{2} - 1 \\ \lambda_{2k} = \lambda_k + 1 & \text{for } 1 \leq k \leq \frac{p}{2} - 1. \end{cases} \quad (1)$$

For example, Figure 2 shows the two Hamiltonian paths HP_{00000} in LTQ_5^0 and HP_{00001} in LTQ_5^1 , respectively. In this figure, two nodes x_i and $x_{i'}$ joining an edge with label λ means that $N_\lambda(x_i) = x_{i'}$ (or $N_\lambda(x_{i'}) = x_i$). Moreover, the additional 4-dimensional edges shown by grey lines are caused by the following lemma.

Lemma 3 *For $n \geq 3$ and $j \in \{0, 1\}$, let $x \in LTQ_n^j$ be any node and $p = 2^{n-1}$. If $HP_x =$*

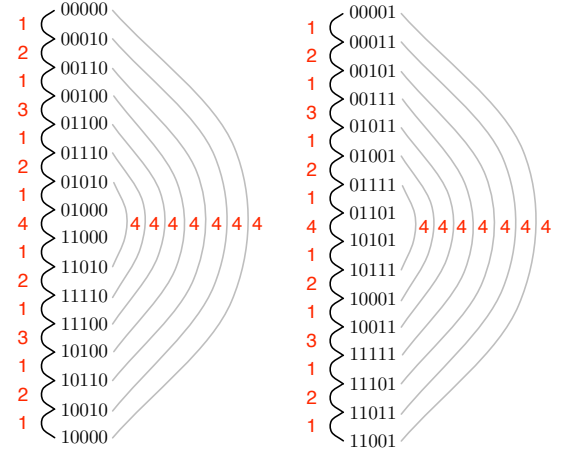


Figure 2: Two particular Hamiltonian paths in the subgraphs LTQ_5^0 and LTQ_5^1 .

($x = x_1, x_2, \dots, x_p$), then $N_{n-1}(x_{p-k+1}) = x_k$ for $1 \leq k \leq p$.

Proof. By symmetry, without loss of generality we assume $k < p-k+1$. Suppose that (x_i, x_{i+1}) is an λ_i -dimensional edge. By (1), it is easy to check

$$\begin{aligned} \lambda_{p/2} &= \lambda_{2^{n-2}} = \lambda_{2^{n-3}} + 1 = \lambda_{2^{n-4}} + 2 = \dots \\ &= \lambda_1 + (n-2) = n-1. \end{aligned}$$

Since $(x_k, x_{k+1}, \dots, x_{p-k+1})$ is the subpath of HP_x , we have

$$x_k = N_{\lambda_k}(N_{\lambda_{k+1}}(\dots N_{\lambda_{p-k}}(x_{p-k+1}) \dots)).$$

By Proposition 1 and the fact $\lambda_i = \lambda_{p-i}$ for all $k \leq i \leq p-k$, we obtain

$$\begin{aligned} x_k &= N_{\lambda_k}(N_{\lambda_{k+1}}(\dots N_{\lambda_{p/2-1}}(N_{\lambda_{p/2}}(N_{\lambda_{p/2-1}}(\dots \\ &\quad \dots N_{\lambda_{k+1}}(N_{\lambda_k}(x_{p-k+1})) \dots))) \dots)) \\ &= N_{\lambda_k}(N_{\lambda_k}(N_{\lambda_{k+1}}(N_{\lambda_{k+1}}(\dots N_{\lambda_{p/2-1}}(N_{\lambda_{p/2-1}}(N_{\lambda_{p/2}}(x_{p-k+1})) \dots))) \dots)) \\ &= N_{\lambda_{p/2}}(x_{p-k+1}) \\ &= N_{n-1}(x_{p-k+1}). \end{aligned} \quad \square$$

Indeed, the result of Lemma 3 can be inherited to subpaths of HP_x . To show this, we need the following notation. Let ${}^{(\ell)}LTQ_n$ denote an induced subgraph of LTQ_n whose nodes are fixed with a certain prefix label of length ℓ . In particular, ${}^{(\ell)}LTQ_n^j$ is the subgraph of ${}^{(\ell)}LTQ_n$

obtained by suffixing j as the last bit in each node. Let $E_i(LTQ_n)$ denote the edge set consisting of all i -dimensional edges in LTQ_n . Obviously, $|E_i(LTQ_n)| = 2^{n-1}$ for all $0 \leq i \leq n-1$. From the recursive structure of HP_x , the following lemma can be proved by a similar argument as that in Lemma 3. For example, Figure 3 demonstrates the result.

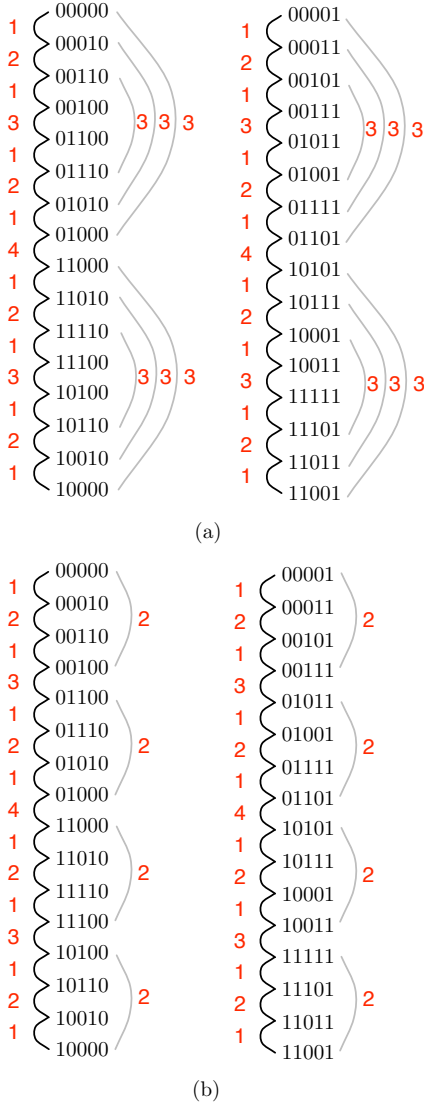


Figure 3: 3-dimensional edges and 2-dimensional edges in LTQ_5^0 and LTQ_5^1 .

Lemma 4 For $n \geq 3$ and $j \in \{0, 1\}$, let $x \in {}^{(\ell)}LTQ_n^j$ be any node and $p = 2^{n-\ell-1}$. If $HP_x = (x = x_1, x_2, \dots, x_p)$, then $(x_k, x_{p-k+1}) \in E_{n-\ell-1}(LTQ_n)$ for $1 \leq k \leq p$.

We close this section with giving the definition of multi-dimensional meshes. A k -dimensional mesh (k -D mesh for short) of size $\ell_1 \times \ell_2 \times \dots \times \ell_k$, denoted by $M_{\ell_1 \times \ell_2 \times \dots \times \ell_k}$, is a graph with node set $\{(m_1, m_2, \dots, m_k) \mid 1 \leq m_i \leq \ell_i \text{ for } 1 \leq i \leq k\}$ such that every node $(m_1, m_2, \dots, m_k)$ is connected to nodes $(m_1, m_2, \dots, m_i \pm 1, \dots, m_k)$ for $1 \leq i \leq k$ provided they exist (i.e., $1 \leq m_i \pm 1 \leq \ell_i$). For simplicity, we write $m_1 m_2 \dots m_k$ instead of $(m_1, m_2, \dots, m_k)$ to represent a node. In particular, a 2-D mesh can be represented by a matrix $M_{\ell_1 \times \ell_2} = (\alpha_{i,j})_{\ell_1 \times \ell_2}$, where node set $V(M_{\ell_1 \times \ell_2}) = \{\alpha_{i,j} \mid 1 \leq i \leq \ell_1 \text{ and } 1 \leq j \leq \ell_2\}$ and edge set $E(M_{\ell_1 \times \ell_2}) = \{(\alpha_{i,j}, \alpha_{i,j+1}) \mid 1 \leq i \leq \ell_1 \text{ and } 1 \leq j \leq \ell_2 - 1\} \cup \{(\alpha_{i,j}, \alpha_{i+1,j}) \mid 1 \leq i \leq \ell_1 - 1 \text{ and } 1 \leq j \leq \ell_2\}$. For $M_{2,\ell}$, the edge $(\alpha_{1,j}, \alpha_{2,j})$ for $1 \leq j \leq \ell$ is called the j th ladder-edge. In addition, the first ladder-edge and the last ladder-edge are also called two side-edges of $M_{2,\ell}$.

3 Main results

Lemma 5 If $e_1, e_2 \in E_2(LTQ_3)$ are two distinct edges lying on a 4-cycle, then there exist two disjoint meshes M and M' each of size 2×2 contained in LTQ_3 such that e_1 and e_2 are two side-edges of M or M' .

Proof. Note that $E_2(LTQ_3) = \{(000, 100), (010, 110), (001, 111), (011, 101)\}$. Let $e_1, e_2 \in E_2(LTQ_3)$ be two distinct edges in a 4-cycle of LTQ_3 . Then, $\{e_1, e_2\} = \{(000, 100), (010, 110)\}$ or $\{e_1, e_2\} = \{(001, 111), (011, 101)\}$. Let M and M' be two 2×2 meshes in LTQ_3 given as follows:

$$M = \begin{pmatrix} 000 & 010 \\ 100 & 110 \end{pmatrix} \text{ and } M' = \begin{pmatrix} 001 & 011 \\ 111 & 101 \end{pmatrix}.$$

It is easy to see that e_1 and e_2 are two side-edges of M or M' . $\square$

The following lemma is a generalization of Lemma 5.

Lemma 6 For $n \geq 3$ and $j \in \{0, 1\}$, if (x, y, z, w, x) is a 4-cycle in LTQ_n^j where $(x, y), (z, w) \in E_{n-1}(LTQ_n)$ and $(y, z), (w, x) \in E_{n-2}(LTQ_n)$, then there exists a mesh M of size $2 \times 2^{n-2}$ contained in LTQ_n^j such that (x, y) and (z, w) are two side-edges of M .

Proof. Since y is the $(n-1)$ -neighbor of x , by Lemma 2 we can construct the particular Hamiltonian path HP_x that connects x and y in LTQ_n^j .

We denote $HP_x = (x = x_1, x_2, \dots, x_p = y)$ where $p = 2^{n-1}$ and suppose that (x_i, x_{i+1}) is an λ_i -dimensional edge for $1 \leq i \leq p-1$. Since $w = N_{n-2}(x)$ and $z = N_{n-2}(y)$, from the constructing rule of HP_x we know that $w = x_{p/2}$ and $z = x_{p/2+1}$. Moreover, $(x = x_1, x_2, \dots, x_{p/2} = w)$ and $(y = x_p, x_{p-1}, \dots, x_{p/2+1} = z)$ are two subpaths each of length 2^{n-2} in HP_x . By Lemma 3, $N_{n-1}(x_k) = x_{p-k+1}$ for $1 \leq k \leq p/2$. This shows that LTQ_n^j contains the following mesh of size $2 \times 2^{n-2}$:

$$M = \begin{pmatrix} x=x_1 & x_2=N_{\lambda_1}(x_1) & x_3=N_{\lambda_2}(x_2) & \dots & x_{\frac{p}{2}}=N_{\lambda_{\frac{p}{2}-1}}(x_{\frac{p}{2}-1})=w \\ y=x_p & x_{p-1}=N_{\lambda_{p-2}}(x_{p-2}) & x_{p-2}=N_{\lambda_{p-3}}(x_{p-3}) & \dots & x_{\frac{p}{2}+1}=N_{\lambda_{\frac{p}{2}}}(x_{\frac{p}{2}})=z \end{pmatrix}.$$

It is easy to see that (x, y) and (z, w) are two side-edges of M . $\square$

Corollary 7 For $n \geq 3$, LTQ_n contains two disjoint meshes each of size $2 \times 2^{n-2}$, where one mesh is contained in LTQ_n^0 and the other is contained in LTQ_n^1 .

We now prove the main result as follows.

Theorem 8 For $n \geq 3$ and $2 \leq k \leq n$, a family of two disjoint meshes each of size $\underbrace{2 \times 2 \times \dots \times 2}_{k-1} \times 2^{n-k}$ can be embedded into LTQ_n with unit dilation, unit expansion, and congestion-free. In particular, the mesh of size $\underbrace{2 \times 2 \times \dots \times 2}_{k-1} \times 2^{n-k}$ is isomorphic to Q_{n-1} if $k = n$ or $k = n-1$.

Proof. For $k = 2$, we have shown by Corollary 7 that there exist two disjoint meshes each of size $2 \times 2^{n-2}$ in LTQ_n . In particular, the construction of such a mesh in LTQ_n^j for $j \in \{0, 1\}$ relies on a particular Hamiltonian path HP_x where $x \in LTQ_n^j$. In the following proof, we only consider the embedding of a mesh of size $\underbrace{2 \times 2 \times \dots \times 2}_{k-1} \times 2^{n-k}$ in LTQ_n^0 for $3 \leq k \leq n$, and omit the case of embedding in LTQ_n^1 since it can be done by a similar way. For any node $x \in LTQ_n^0$, we first divide HP_x into 2^{k-2} subpaths with equal length. Let $p = 2^{n-k+1}$ be such a length and denote $Q_i = (x_{i,1}, x_{i,2}, \dots, x_{i,p})$ for $1 \leq i \leq 2^{k-2}$ as those paths. From the recursive structure of HP_x , nodes in each Q_i is obviously contained in a graph ${}^{(\ell)}LTQ_n^0$, where $\ell = k-2$ is the length of a certain prefix label. Since ${}^{(\ell)}LTQ_n^0$

is isomorphic to $LTQ_{n-\ell}^0$, by Lemma 6 ${}^{(\ell)}LTQ_n^0$ contains a mesh of size $2 \times 2^{n-k}$ as follows:

$$M_i = \begin{pmatrix} x_{i,1} & x_{i,2} & \dots & x_{i,\frac{p}{2}} \\ x_{i,p} & x_{i,p-1} & \dots & x_{i,\frac{p}{2}+1} \end{pmatrix}.$$

Clearly, the mesh takes $(x_{i,1}, x_{i,p})$ and $(x_{i,p/2}, x_{i,p/2+1})$ as the side-edges.

We now show how to embed a high dimensional mesh in LTQ_n^0 . For two meshes M_i and M_{i+1} where i is an odd integer, since Q_{i+1} follows Q_i in HP_x , by Lemma 4 we know that two nodes $x_{i,s} \in Q_i$ and $x_{i+1,t} \in Q_{i+1}$ with $s+t = 1+p$ are connected by an $(n-\ell)$ -dimensional edge. Thus, combining this two meshes together with p $(n-\ell)$ -dimensional edges, the resulting graph forms a mesh of size $2 \times 2 \times 2^{n-k}$. As a consequence, LTQ_n^0 contains $2^{\ell-1}$ such kind of meshes. By a similar procedure that combines two meshes each of size $2 \times 2 \times 2^{n-k}$, we can obtain a mesh of size $2 \times 2 \times 2 \times 2^{n-k}$. In general, let M_i , $1 \leq i \leq 2^{k-d-1}$, be meshes each of size $\underbrace{2 \times 2 \times \dots \times 2}_d \times 2^{n-k}$ for $1 \leq d \leq k-2$

such that nodes in each mesh M_i , denoted by $y_{i,1}, y_{i,2}, \dots, y_{i,q}$ where $q = 2^{n-k+d}$, consecutively appear in HP_x . Moreover, $M_1, M_2, \dots, M_{k-d-1}$ appear in series. Consider two meshes M_i and M_{i+1} where i is an odd integer. By Lemma 4, $(y_{i,s}, y_{i+1,t}) \in E_{n-k+d+1}(LTQ_n)$ provided $s+t = 1+q$. Thus, combining M_i and M_{i+1} together with q $(n-k+d+1)$ -dimensional edges, we obtain the desired mesh of size $\underbrace{2 \times 2 \times \dots \times 2}_{d+1} \times 2^{n-k}$. $\square$

4 Concluding remarks

This paper has investigated how to embed a family of disjoint multi-dimensional meshes into locally twisted cubes. We prove that for $n \geq 3$ and $2 \leq k \leq n$, a family of two disjoint meshes each of size $\underbrace{2 \times 2 \times \dots \times 2}_{k-1} \times 2^{n-k}$ can be embed-

ded into LTQ_n with unit dilation, unit expansion, and congestion-free. To the best of our knowledge, it seems that the embedding appeared here is the first time to consider disjoint meshes as guest graphs for embedding into locally twisted cubes. For the future study, it is also worthwhile to investigate how to embed a family of disjoint multi-dimensional meshes with different sizes.

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